

Journal Pre-proofs

Research Paper

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PII: S1674-9871(26)00121-0
DOI: <https://doi.org/10.1016/j.gsf.2026.102367>
Reference: GSF 102367

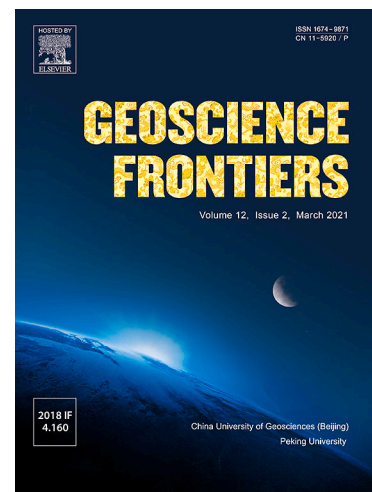
To appear in: *Geoscience Frontiers*

Received Date: 6 January 2026
Revised Date: 29 March 2026
Accepted Date: 1 June 2026

Please cite this article as: Y. Cui, M. Usman, M. Irfan, M. Haseeb, Artificial intelligence and natural resource exploitation shaping load capacity factor in Quad economies: Role of renewable energy and green innovations, *Geoscience Frontiers* (2026), doi: <https://doi.org/10.1016/j.gsf.2026.102367>

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Research Paper

Artificial intelligence and natural resource exploitation shaping load capacity factor in Quad economies: Role of renewable energy and green innovations

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Abstract

This study examines the heterogeneous effects of artificial intelligence (AI) adoption and natural resource exploitation on load capacity factor (LCF) in the Quad economies over the period 1995–2023. After accounting for cross-sectional dependence (CD) and slope heterogeneity (SH), second-generation econometric methods are employed to estimate reliable outcomes. To capture variation across different levels of ecological capacity, the Method of Moments Quantile Regression (MMQR) is applied. The estimated findings reveal that AI adoption has a statistically significant and negative influence on LCF, with coefficients declining from -0.240 at lower quantiles to -0.305 at upper quantiles, indicating stronger adverse environmental effects at higher levels of ecological capacity. In contrast, natural resources exert a positive influence on LCF, with effects strengthening from 0.305 to 0.330 across the quantile distribution. Also, green innovation contributes positively to environmental sustainability, with its impact rising from 0.093 at lower quantiles to 0.249 at higher quantiles. Financial development emerges as the most influential

determinant of LCF, as indicated by the largest coefficient magnitudes across quantiles, with values ranging from 1.033 to 1.406. Additionally, renewable energy displays a stronger positive relation with LCF, growing from 0.421 at the lower quantile to 0.764 at the upper quantile. These results highlight the importance of integrated strategies combining AI governance, sustainable resource management, financial expansion, and renewable energy development to enhance environmental sustainability in the Quad economies.

Keywords: Load capacity factor; Artificial intelligence; Natural resources; Renewable energy; Quad economies

1. Introduction

Environmental degradation directly impacts public health and ecosystem stability. Diminishing water and air quality, exposure to pollutants, and limited access to hygienic drinking water contribute to respiratory illnesses, cancer, and other health issues. Ecosystem disruptions, including deforestation, habitat loss, and biodiversity decline, undermine ecological resilience and threaten sustainable resource use (Chen et al., 2025). Global warming and climate change are among the most persistent worldwide contests, bringing extreme weather events, food insecurity, biodiversity loss, and economic risks, which necessitate a transition to greener economies. The IPCC (2018) projects global temperatures could rise by 1.5 °C between 2030 and 2052 if current trends continue. Anthropogenic greenhouse gas (GHG) emissions, mainly from fossil fuel combustion and deforestation, exacerbate these effects, contributing to extreme weather events, rising sea levels, and other ecological impacts. Between 1990 and 2022, total global greenhouse gas emissions increased by 61.7%, though the COVID-19 pandemic temporarily reduced emissions by 3.7% in 2020. Emissions have since resumed their upward trajectory, with a 2.3% increase from 2019 and a 1.4% rise from 2021. Per capita emissions also rose by 8.3%, from 6.24 to 6.76 t CO₂ (Kartal et al., 2025). Addressing global warming necessitates coordinated comprehensive efforts to diminish environmental pollution, evolution to renewable energy, restore ecosystems, and implement adaptive strategies to safeguard environmental and societal well-being.

The global economy is experiencing a reflective transformation driven by rapid advances in artificial intelligence (AI), which have reshaped production systems, governance structures, and social interactions. The AI adoption across sectors like healthcare, finance, transportation, manufacturing, logistics, and public administration has significantly enhanced efficiency, productivity, innovation, and long-term economic growth. Nevertheless, the swift diffusion of AI has also elevated increasing ecological concerns (Ahmed et al., 2025). These concerns largely stem from the intensive computational necessities of progressive AI models, which depend on energy-intensive data centers and incessant data processing (Strubell et al., 2019). In many countries, this demand of energy is still met primarily through nonrenewable-based sources, contributing to higher environmental pressure (Wang et al., 2025). Additionally, the development of digital infrastructure, incorporating cloud platforms and large-scale data storage systems, increases resource extraction and electronic waste generation (Niu et al., 2025). Simultaneously, AI holds strong potential to enhance environmental sustainability by improving energy efficiency, optimizing resource allocation, and curtailing environmental pollution across industrial and

transportation systems (Fan et al., 2025). This dual role highlights the significance of aligning AI-driven innovation with environmental sustainability goals.

Moreover, natural resources play a vital role in shaping the load capacity factor (LCF) by influencing both environmental demand and supply dynamics. An economy's capital accumulation is often driven by the natural resource extraction such as coal, oil, minerals, timber, and other raw materials, which generate substantial resource rents (Wang and Yuan, 2026). Though, when these rents are largely used to sustain current consumption instead of being reinvested in productive capital, clean technologies, and infrastructure, environmental degradation intensifies and long-term sustainability is compromised. Industrialization further accelerates the exploitation and extraction of natural resources, leading to reductions in biocapacity and increases in ecological footprint, thereby exerting negative pressure on LCF and resulting in ecological deficits (Ahmad et al., 2020). Between 1980 and the present, developing economies have generated approximately USD 24.72 trillion in natural resource rents, underscoring the scale of this challenge. The association between environmental quality and natural resources remains contentious, as resource abundance can stimulate economic growth while simultaneously exacerbating pollution through intensified extraction (Sun et al., 2024). Ecological footprint indicators capture these pressures by accounting for human-induced demands on land, forests, grazing areas, cropland, marine ecosystems, built-up land, and carbon emissions, offering a comprehensive measure of environmental stress linked to natural resource use.

Since the early 1990s, global environmental initiatives have increasingly emphasized the transition toward low- and zero-carbon production systems. Landmark contracts like the Kyoto Protocol (1997) and the Paris Agreement (2015) have underscored the critical role of green technological innovation (GTI) in mitigating climate change. Although countries differ in their emission-reduction targets and implementation strategies, the sustainable diminishing of environmental pollution largely depends on the GTI adoption. These technologies enhance production efficiency while simultaneously lowering emission intensity, thereby improving industrial performance and environmental outcomes. Moreover, GTI contributes to diminishing environmental degradation by enabling cleaner production processes and creating opportunities to curb greenhouse gas (GHG) emissions (Hussain and Dogan, 2021). By improving production efficiency, GTI can also increase export competitiveness, potentially shifting pollution-intensive production away from domestic economies and reducing local environmental pressure. This mechanism is particularly relevant for the Quad economies, which cooperatively account for an extensive share of global trade and industrial output. Consequently, examining the association between GTI and ecological sustainability in the Quad context is both timely and policy-relevant, given their pivotal role in global emissions reduction and technological leadership.

Renewable energy sources, unlike non-renewable alternatives, offer substantial environmental benefits but typically require significant upfront investment before yielding long-term returns. The expansion of clean and alternative energy resources, such as geothermal, wave, hydropower, solar, wind and biomass energy represent a critical pathway toward sustainable economic growth. Consequently, worldwide economies are increasingly investing in renewable and cleaner energy to reduce ecological pressures and boost energy security (Kartal et al., 2024a, 2024b). High-income economies, in particular, have accelerated the transformation of their industrial structures

toward renewable energy-based systems and are more advanced in this transition process. While existing studies have examined the influence of renewable energy consumption on the LCF in APEC countries (Sun et al., 2024), G7 countries (Kartal, 2024), and selected 8 countries (Ozcan et al., 2023), empirical evidence for the Quad economies remains scarce. Therefore, examining the role of renewable energy in improving LCF within the Quad context provides a valuable contribution to the existing literature and provides important policy insights for economies that play an essential role in global energy transitions and environmental sustainability.

Focusing on the Quad economies is predominantly important due to their extensive contribution to global greenhouse gas (GHG) emissions, diverse economic structures, and significant influence on regional and global climate governance. The Quad economies, including three advanced industrialized economies, the United States, Australia, and Japan, and one rapidly emerging economy, India represents a unique combination of technological capacity, industrial output, and energy demand, making it a critical setting for inspecting the interactions between AI, natural resources, energy use, and environmental sustainability. Economic interdependencies among these countries, such as trade, supply chains, financial development, and resource flows, amplify the cross-border implications of energy use and GHG emissions. Over the past decade, the Quad economies have shown a strong obligation to climate action through initiatives for example the Quad Climate Change Adaptation and Mitigation Package (Q-CHAMP) launched in 2022, exertions to reinforce green energy supply chains (Quad Leaders' Joint Statement, 2023), and substantial financial support for renewable energy and sustainable infrastructure projects. These measures aim to boost environmental resilience in the Indo-Pacific region, promote data-driven environmental monitoring, and advance green technological innovation. By coordinating policies and investments, the Quad nations align with global frameworks for instance the Glasgow Climate Pact and the Paris Agreement, leveraging shared values, rule-of-law commitments, and multilateral cooperation to reduce emissions, foster sustainable development, and provide a model for international collaboration that transcends traditional Global North–South divides (Mahajan and Sah, 2026).

Within the Sustainable Development Goals (SDGs) framework, researchers have increasingly sought more comprehensive indicators to capture the multifaceted nature of environmental degradation. Early empirical studies predominantly relied on carbon dioxide (CO₂) emissions as a proxy for environmental quality (Xiaoman et al., 2021; Li et al., 2022). While CO₂ emissions effectively reflect energy-related pollution, they offer only a partial assessment of environmental degradation, as they fail to capture non-energy-related pressures such as land use change, resource depletion, and ecosystem degradation. Consequently, scholars have expanded the scope of environmental indicators to include ecological footprints (EFP) (Chen et al., 2025; Yu et al., 2025), GHG emissions (Haseeb et al., 2025; Usman et al., 2025), and the environmental performance index (Iqbal et al., 2021), each offering broader insights into environmental stress. Among these measures, the ecological footprint has gained prominence because it accounts for multiple dimensions of environmental pressure, including energy consumption, land use, and resource extraction. The EFP, developed by Rees (1992), captures anthropogenetic compressions on the atmosphere by accounting for the impacts of human activities on water, air, and soil systems. This accounting is structured around two key dimensions: demand-side ecological footprint and supply-side biocapacity (Galli, 2015). While the EFP measures the demand placed on natural resource

reserves by human consumption and production activities, biocapacity signifies the Earth's capability to restore these resources and fascinate waste. The interaction between these two dimensions provides a comprehensive assessment of environmental sustainability, highlighting whether natural systems can meet human demands without ecological degradation. However, EFP remains insufficient in that its attentions exclusively on the demand side of ecological services, neglecting the Earth's capacity to restore resources and absorb waste. This omission can lead to an incomplete assessment of environmental sustainability, particularly in the framework of long-term ecological balance. To address this limitation, biocapacity has been introduced as a complementary indicator, representing the Earth's capacity to supply biological resources and assimilate waste. Building on this conceptual advancement, Siche et al. (2010) anticipated the LCF, well-defined as the ratio of biocapacity to ecological footprint. The LCF provides a holistic measure of environmental sustainability by jointly capturing environmental demand and supply. An LCF value over and above one shows ecological sustainability, values lower one signal environmental overshoot, and a value equal to one represents a critical threshold. Recent studies have increasingly adopted LCF as a superior indicator of environmental quality, as it offers a more integrated and policy-relevant assessment of sustainability (Sun et al., 2024). Fig. 1 exhibits the notable differences of the LCF across the Quad economies during 1995–2023. In this regard, Australia consistently recorded the highest LCF values, although its performance declined from 2.5029 in 1995 to 1.6347 in 2023. However, the United States showed a relatively stable trend, increasing from 0.4357 in 1995 to 0.4915 in 2023, with a peak of 0.5278 in 2020. Moreover, Japan experienced a gradual improvement from 0.1140 in 1995 to 0.1539 in 2023. In contrast, India maintained the lowest LCF levels throughout the study period, declining from 0.4739 in 1995 to 0.2962 in 2023. These findings indicate substantial disparities in ecological sustainability among the Quad economies, with Australia maintaining a significant ecological surplus despite a downward trend.

In conclusion, the present study makes a meaningful and incremental contribution to the existing literature by addressing several critical research gaps related to environmental sustainability in the Quad economies. First, this study systematically examines the roles of artificial intelligence and natural resources in shaping the LCF, while simultaneously accounting for green technological innovation, financial development, and renewable energy use within the Stochastic Impacts by Regression on Population, Affluence, and Technology (STIRPAT) model framework. While previous research has examined AI and natural resource utilization separately in environmental contexts, their combined and conditional effects on ecological capacity remain largely unexplored, predominantly in the context of advanced and emerging Quad countries. Second, the existing literature offers limited empirical evidence on the key determinants of the LCF as an inclusive measure of environmental quality. Studies focusing on the Quad economies is particularly scarce, despite their significant economic size, technological leadership, and ecological footprint. The motivation for concentrating on the Quad countries lies in their deep economic interconnections in today's globalized world, encompassing trade, and foreign direct investment, and supply chains. Economic activities in one country, such as coal exports from Australia to India and Japan can have direct implications for the production and energy consumption patterns of partner countries. Moreover, it is well established that economic activity is closely linked to carbon emissions. Given the strong economic integration among the Quad nations and their shared development and sustainability objectives, there is substantial reason to expect that their carbon emissions are likely

to exhibit long-run cointegration. It is also imperative to note that the energy composition of the Quad countries has remained largely stable over the past two decades. Fossil fuels continue to account for approximately 40%–50% of their energy mix, with crude oil remaining central to economic activity. Consequently, these economies rely on similar carbon-intensive energy resources, which are significant contributors to environmental pollution. This shared dependence on polluting energy sources reinforces the argument for analyzing the Quad countries collectively when examining carbon emissions and environmental interdependencies. Moreover, this study addresses a critical gap in the literature by explicitly scrutinizing the influence of artificial intelligence, natural resource rents, green technological innovations, and renewable energy adoption on environmental sustainability. It provides novel insights into how growth-oriented economies can strengthen their ecological capacity while remaining within environmental limits, highlighting pathways for sustainable development that integrate technological advancement with environmental stewardship. Third, this study applies advanced second-generation econometric procedures that explicitly account for slope heterogeneity and cross-sectional dependence, thereby improving the reliability and robustness of the estimated relationships. Finally, the novel outcomes emphasize the importance of coordinated adoption of AI, green technological innovations, sustainable natural resource management, financial expansion, and renewable energy deployment in the Quad economies. Such an integrated strategy is important for accomplishing long-term economic development while safeguarding ecological sustainability and enhancing the load capacity factor across these economies.

2. Theoretical framework and model

The artificial intelligence can influence environmental quality in the Quad economies through multiple technological and economic channels. From the view of the STIRPAT Model, AI indicates a critical technological factor that can both mitigate and exacerbate environmental pressures. On the one hand, AI-driven innovations enhance environmental quality by improving energy productivity, empowering smart grid management, augmenting industrial production processes, and facilitating resource-efficient systems in transportation, agriculture, and manufacturing sectors. These applications support the transition toward low-carbon development by promoting renewable energy integration, reducing waste, and enabling data-driven environmental monitoring. On the other hand, the rapid expansion of AI infrastructure, including high-performance computing structures, data centers, and digital automation can significantly increase electricity consumption and carbon emissions if powered by fossil fuel-based power systems. Consequently, the net environmental consequence of AI in Quad economies depends on the extent to which AI development is integrated with green technological innovation, renewable energy adoption, and effective environmental regulations. When supported by sustainable energy frameworks and green policy incentives, AI can act as a catalyst for improving environmental quality and strengthening ecological sustainability across technologically advanced and rapidly developing economies.

The fundamental econometric background of this article is grounded in multiple linear regression models designed to scrutinize the influence of AI technologies and other explanatory indicators on the LCF in the Quad economies. The theoretical foundation originates from the well-known IPAT identity ($I = P \times A \times T$), proposed by Ehrlich and Holdren (1971), which conceptualizes ecological

influence as a multiplicative function of population size, affluence, and technology. To overcome the deterministic limitations of the original IPAT formulation, Dietz and Rosa (1994) later introduced its stochastic extension, known as the STIRPAT (Stochastic Impacts by Regression on Population, Affluence, and Technology) model, which allows for empirical estimation and hypothesis testing. The IPAT framework implies that environmental pressure arises from the combined effects of demographic expansion, economic activity, and technological progress. In this study, the model is innovatively adapted to capture the association between AI technologies and ecological sustainability, as measured by the LCF. A multiple linear regression approach is employed because it enables the flexible estimation of individual effects while avoiding the restrictive assumptions of fixed proportionality embedded in the original IPAT identity. Specifically, environmental quality (I) is modeled as a function of population dynamics (P), affluence demonstrating the level of economic activity (A), and technological development (T), represented by AI technology and related indicators. To assess how these factors influence the LCF, the empirical strategy isolates the effect of each component while holding the others constant. This transformation consents for a more nuanced understanding of how changes in technology or economic conditions affect ecological capacity. Therefore, the extended IPAT/STIRPAT framework is formally stated in Eq. (1) as follows:

$$I_{it} = MP_{it}^{\alpha} A_{it}^{\beta} T_{it}^{\gamma} \mu_{it} \quad (1)$$

To relax the restrictive assumption of constant proportionality embedded in the original IPAT context, the model incorporates α , β , and γ as estimable parameters associated with population (P), affluence (A), and technology (T), respectively. This specification allows the elasticities of environmental impact with respect to each component to vary across economies and over time. In addition, M signifies the constant term, while μ represents the stochastic error component capturing unobserved influences. The subscripts i and t denote to the cross-sectional units and the time dimension, respectively. By applying a logarithmic transformation, the multiplicative IPAT relationship is converted into a linear and empirically tractable form. Accordingly, the log-linear specification of Eq. (1) is expressed in Eq. (2) as follows:

$$\ln(I_{it}) = M_i + \alpha(\ln P_{it}) + \beta(\ln A_{it}) + \gamma(\ln T_{it}) + \mu_{it} \quad (2)$$

The IPAT relationship is specified in logarithmic (elasticity) form to facilitate empirical estimation. Within this framework, the study aligns with the established literature by employing the I = PAT structure to examine environmental impacts in a unified and consistent manner. Accordingly, the empirical model is formulated in a mathematical form and presented in Eq. (3) as follows:

$$\ln(LCF_{it}) = \beta_0 + \beta_1 \ln(AI_{it}) + \beta_2 \ln(NR_{it}) + \beta_3 \ln(GTI_{it}) + \beta_4 \ln(FD_{it}) + \beta_5 \ln(REC_{it}) + \mu_{it} \quad (3)$$

Here, (t) signifies time periods and (i) denotes the cross-sectional units. The symbol β_0 captures country-specific fixed effects, μ_{it} is the error term, and $(\ln)$ designates natural logarithms. The parameters $\beta_1 - \beta_5$ represent the elasticities of all explanatory variables.

3. Data description and methodology

3.1 Data description and measurement

This study employs annual country-level data from internationally recognized sources to examine the drivers of environmental sustainability. Environmental quality is measured by the load capacity factor (LCF), delineated as the ratio of biocapacity per capita to ecological footprint per capita (global hectares per person, GHAp), sourced from the Global Footprint Network (GFPN, 2024). Artificial intelligence (AI) development is proxied by the number of AI patent applications filed under the Patent Cooperation Treaty (PCT), obtained from the World Intellectual Property Organization (WIPO, 2024). Green technological innovation (GTI) is measured by total patents, taken from the OECD database (OECD, 2024). Financial development (FD) is captured using the IMF's financial development index, reflecting financial market and institutional access, depth, and efficiency (IMF, 2024). Natural resource (NR) rents, expressed as a percentage of GDP, and renewable energy consumption (REC), as a share of total final energy use, are extracted from the World Development Indicators (WDI, 2024). These indicators provide an inclusive background to investigate the interactions between technological progress, financial progress, resource utilization, and environmental sustainability. Accordingly, the present study explores the influence of AI, NR, GTI, FD, and REC on LCF across Quad economies.

The descriptive analysis provides a preliminary overview of the variability, behavior, and trends of the important variables employed in this study across the Quad economies. In this way, Fig. 2 illustrates the long-term evolution of AI development. The results reveal a clear upward trajectory in AI adoption across the Quad countries. The United States records the most substantial growth (particularly after 2010), reflecting its global technological leadership and significant investments in digital infrastructure and innovation. Japan also demonstrates relatively high levels of AI development, particularly during the early years of the sample period, while Australia exhibits moderate but steady progress. In contrast, India maintains comparatively lower levels of AI development, although a gradual upward trend is evident over time. Moreover, Fig. 3 presents the temporal dynamics and comparative evolution of total natural resource rents as a percentage of GDP across the Quad economies. The results indicate considerable cross-country variation. Australia records the highest and most volatile resource rents, reflecting its strong reliance on mineral and energy exports. India shows moderate and relatively stable levels of resource rents, whereas the United States experiences moderate fluctuations over the study period. In contrast, Japan exhibits minimal values due to its limited domestic natural resource endowment. In addition, Fig. 4 depicts the evolution of green technological innovation, measured by environmental-related patent activity from 1995 to 2023. The results suggest a general upward trend in green technological innovation across most countries. The United States leads with the highest level of growth, followed by Japan, which also maintains a strong capacity for green technological innovation. Australia displays moderate fluctuations, while India shows gradual improvement, although its overall level remains comparatively lower. These upward trajectories reflect increasing commitments to sustainable innovation, environmental protection, and the development of cleaner production technologies across the Quad economies.

In Fig. 5, the financial development index improved steadily across the Quad countries over the study period. The United States maintained the highest financial development levels, with index

values around 1 in recent years, indicating highly developed financial institutions and markets. Japan also demonstrates consistent progress, whereas India begins from a relatively lower base but gradually strengthens its financial sector during the study period. These trends reflect ongoing financial sector reforms, greater market depth, enhanced accessibility, and improved efficiency of financial systems across the Quad economies. Finally, Fig. 6 illustrates the trends in renewable energy consumption across the Quad economies from 1995 to 2023. The results reveal considerable cross-country variation in the share of renewable energy use. India consistently recorded the highest levels of renewable energy consumption, exceeding 50% of total final energy use throughout several years of the study period. Australia also maintained a relatively high share of renewable energy consumption and exhibited a gradual upward trend over time. In contrast, the United States and Japan reported comparatively lower levels of renewable energy use, although both countries experienced steady improvements, particularly after 2010. This increasing dependence on renewable energy across the Quad economies reflects ongoing efforts to promote energy transition, reduce environmental pressures, and advance sustainable development objectives.

Fig. 7 presents the quantile distribution plots for the study variables, including LCF, AI, NR, GTI, FD, and REC. The plots compare the observed quantiles of each variable with the theoretical quantiles of a normal distribution. Overall, most observations cluster closely around the reference line in the central quantiles, suggesting that the variables are around normally distributed in their mid-ranges. However, noticeable deviations occur in the lower and upper tails for several variables, indicating the presence of mild skewness and potential outliers. In particular, AI, GTI, and REC exhibit stronger deviations in the upper quantiles, reflecting positive skewness and suggesting that these variables experienced substantial increases in certain periods of the sample. Similarly, NR and LCF show moderate departures from normality, particularly at the extremes, which may be associated with cross-country heterogeneity and temporal fluctuations in environmental and resource dynamics. In contrast, FD appears relatively closer to the normal distribution with only minor tail deviations, indicating a more stable distribution pattern. These results suggest that while the variables broadly follow normality in their central tendencies, the observed tail deviations highlight the dynamic nature of technological, environmental, and economic developments within the Quad economies over the study period.

3.2 Methodological process

In order to ensure a robust empirical analysis of environmental sustainability in the Quad economies, this study adopts a six-step methodological framework, as illustrated in Fig. 8. The process begins by examining cross-sectional dependence (CD, CD_W , CD_{W+} , CD^*) to identify interlinkages among countries, which allows for the consideration of shared shocks or spillover effects. Recognizing the diversity among these economies, slope heterogeneity tests, such as Pesaran and Yamagata, (2008), and Blomquist and Westerlund, (2013) are employed to account for country-specific variations in the relationships between variables. Once these foundational characteristics are established, panel unit root tests such as Cross-Sectionally Augmented Dickey–Fuller (CADF), and Cross-Sectionally Augmented Im-Pesaran-Shin (CIPS) determine the stationarity of the data, ensuring that subsequent analyses are not compromised by non-stationary series. Furthermore, this study applies panel cointegration tests (Kao, Pedroni, Westerlund) to

detect long-run equilibrium relationships among the variables, confirming whether they move together over time. Building on this foundation, the Method of Moments Quantile Regression (MMQR) is used to capture heterogeneous effects across different quantiles of the load capacity factor, providing a nuanced understanding of how explanatory factors impact countries with varying levels of environmental performance. Finally, the Dumitrescu and Hurlin panel causality test identifies the direction of causal relationships among the variables, revealing how changes in technology, resources, finance, and energy consumption influence environmental outcomes.

3.2.1 Cross-sectional dependence (CD) approaches

The analysis of panel data faces numerous methodological tasks, amongst where CD is predominantly important, as overlooking it can lead to spurious and biased estimates and invalid inferences. Specified the durable economic integration and monetary linkages amongst Quad-Four economies, the existence of CD is highly likely. To account for this issue, this paper employs a variety of advanced CD tests, counting the Pesaran (2015, 2021) CD test, the CD_W test of Juodis and Reese (2022), the CD_{W+} test of Fan et al. (2015), and the CD^* approach proposed by Pesaran and Xie (2021). The analysis mainly relies on the Pesaran (2015, 2021) CD test, this is provided in Eqs. (4) and (5) as follows:

$$CD = \sqrt{\frac{2(T)}{N(N-1)}} \left(\sum_{i=1}^{N-1} \sum_{j=i+1}^N \hat{\lambda}_{ij} \right) \sim N(0,1) \quad i, j \quad (4)$$

$CD = (1, 2, \dots, N)$

$$M = \sqrt{\frac{2(T)}{N(N-1)}} \left(\sum_{i=1}^{N-1} \sum_{j=i+1}^N \hat{\lambda}_{ij} \right) \left[\frac{(T-K)\hat{\lambda}_{ij}^2 - (T-K)\hat{\lambda}_{ij}^2}{Var(T-K)\hat{\lambda}_{ij}^2} \right] \quad (5)$$

The term $\hat{\lambda}_{ij}^2$ denotes the estimated pairwise cross-sectional correlations calculated from data sample, based on residuals obtained from the linear regression model.

$$CD_W = \sqrt{\frac{2}{Tn(n-1)}} \sum_{i=1}^N \sum_{t=2}^T \sum_{j=1}^{i-1} \partial_i \eta_{it} \partial_j \eta_{jt} \quad (6)$$

Juodis and Reese (2022) argue that this approach has restricted power. To overcome this shortcoming, they develop an improved test statistic, building on Fan et al. (2015), by augmenting the spread factor Δ_{nT} to construct the enhanced CD_W statistic, as defined in Eqs. (7) and (8) as follows:

$$CD_{W+} = CD_W + \Delta_{nT} \quad (7)$$

$$\Delta_{nT} = \sum_{i=2}^N \sum_{j=1}^{i-1} |\hat{\theta}_{ij,T}| > 2 \sqrt{\frac{\ln(n)}{T}} \quad (8)$$

From the CD_{W+} technique to be operative, Δ_{nT} must display sufficient variation across both cross-sectional units (N) and time (T), particularly under spatial or network dependence. In addition, Pesaran and Xie (2021) introduce the $CD^*(\hat{\delta}_{nT})$ statistic, which improves robustness in detecting CD, as presented in Eq. (9) as follows:

$$CD^*(\hat{\delta}_{nT}) = CD^* = \frac{CD + \sqrt{\frac{T}{2}}\hat{\delta}_{nT}}{1 - \hat{\delta}_{nT}} \quad (9)$$

The $CD^*(\hat{\delta}_{nT})$ statistic represents the bias-corrected CD approach, and the CD^* method relies on this adjusted measure to enhance the correctness and robustness of detecting CD issue.

3.2.2 Slope heterogeneity (SH) methods

In the analysis of panel data, SH denotes to variations in the associations between explanatory and explained variables across the unit of cross-section (Pesaran and Yamagata, 2008). This heterogeneity arises when the impacts of independent variables differ across countries due to structural, institutional, or contextual factors. Ignoring SH can result in biased parameter estimates and ambiguous statistical inferences (Blomquist and Westerlund, 2013). Therefore, checking for SH is a crucial step in panel data modeling, as it guides the selection of appropriate econometric techniques capable of accommodating cross-country differences. This test statistics are formally specified in Eqs. (10) and (11) as follows:

$$\hat{\Delta}_{SH} = n^{\frac{1}{2}}(2k)^{-\frac{1}{2}}\left(\frac{1}{n}\tilde{y} - k\right) \quad (10)$$

$$\hat{\Delta}_{ASH} = n^{\frac{1}{2}}\left(\frac{2k(t-k-1)}{t+1}\right)^{-\frac{1}{2}}\left(\frac{1}{n}\tilde{y} - 2k\right) \quad (11)$$

Here, $\hat{\Delta}_{SH}$ and $\hat{\Delta}_{ASH}$ represent the delta and adjusted delta SH statistics, respectively. The null hypothesis adopts homogeneous and consistent slope coefficients across cross-sectional components.

3.2.3 Panel unit root tests

Given the verified existence of CD and SH, it becomes necessary to apply unit root approaches and long-run relationship cointegration methods that properly address these issues. In line with this, the study utilizes the CADF test and the CIPS approaches developed by Pesaran (2007) to evaluate the unit root properties of the series. Such approaches are designed to handle cross-sectional heterogeneity as well as common temporal effects, and they are effective in identifying the integration of second-order within the idiosyncratic error mechanisms. The formulation of the CADF test statistics is presented in Eq. (12) as follows:

$$\Delta z_{i,t} = \alpha_i + \beta_i z_{i,t-1} + \gamma_i z_{t-1} + \sum_{j=0}^s d_{ij} \Delta z_{t-j} + \sum_{j=1}^s \pi_{ij} \Delta z_{i,t-j} + \mu_{i,t} \quad (12)$$

Also, based on the CADF test statistics, the CIPS approach can be formally stated as shown in Eq. (13) as follows:

$$\Delta AC_{i,t} = \delta_i + \delta_i X_{i,t-1} + \delta_i \overline{AC}_{t-1} + \sum_{j=0}^p \delta_{ij} \Delta \overline{AC}_{t-j} + \sum_{j=0}^p \delta_{ij} \Delta AC_{i,t-j} + \mu_{it} \quad (13)$$

Here, $\overline{AC}_{t-1}$ and $\Delta \overline{AC}_{t-j}$ denote the cross-sectional averages. Moreover, the CIPS test used in this paper is presented in Eq. (14) as follows:

$$\widehat{CIPS} = \frac{1}{N} \sum_{i=1}^N CADF_i \quad (14)$$

3.2.4 Panel cointegration tests

The presence of variables integrated at different orders allows for the examination of cointegration relationships, which is a necessary condition for applying panel long-run econometric approaches. Second-generation panel cointegration approaches are particularly effective and dependable for identifying long-term relationships between independent and dependent variables, especially when CD and SH exist. In this context, the study applies several cointegration techniques introduced by Pedroni (1999, 2004), Kao (1999), and Westerlund (2005). These methods are based on a null hypothesis (H_0) of the absence of cointegration, while the alternative hypothesis (H_1) suggests the existence of a long-run relationship between the panel series.

Furthermore, this article uses the cointegration approach proposed by Westerlund owing to the potential existence of CD and SH among the variables. When CD occurs across variables and cross-sections, traditional first-generation panel long-run cointegration tests may yield biased and spurious results, leading to unreliable and misleading inferences concerning long-run relationships (Li et al., 2025). To address these concerns, the Westerlund cointegration test is applied. The Westerlund (2005) framework is based on panel-specific autoregressive (AR) statistics as well as common AR statistics, allowing for heterogeneity across cross-sectional units. These statistics are formally expressed in Eqs. (15) and (16) as follows:

$$VR = \sum_{i=1}^N \sum_{t=1}^N \hat{R}_{it}^2 \hat{M}_i^{-1} \quad (15)$$

$$VR = \sum_{i=1}^N \sum_{t=1}^T \hat{R}_{it}^2 \left(\sum_{t=1}^T \hat{M}_i \right)^{-1} \quad (16)$$

where $\hat{R}_{it}^2 = \sum_{j=1}^t \hat{e}_{ij}$, $\hat{M}_i = \sum_{t=1}^T \hat{e}_{it}^2$ and $\hat{e}_{it}^2$ represents the residuals obtained from the panel regression model. Furthermore, VR represents the variance–ratio statistic based on the group mean.

3.2.5 Method of moment quantile regression (MMQR) test

The present paper utilizes the MMQR to analyze the associations among the selected variables, offering a robust framework that captures distributional heterogeneity. Prior to estimation, several diagnostic tests ensure the reliability of MMQR results. The CD tests are first applied to detect interdependencies among variables, as ignoring CD can bias estimates. Moreover, slope heterogeneity tests then assess whether relationships vary across quantiles, a critical consideration in quantile-based models. Also, panel stationary methods are applied to confirm stationarity, followed by cointegration tests to establish the presence of long-run stable equilibrium associations. These preliminary steps provide a solid substance for credible MMQR assessment. This approach extends traditional regression by assessing conditional quantiles in place of concentrating exclusively on the average, allowing the model to capture how regressor effects differ across the outcome distribution (Machado and Silva, 2019). This approach is predominantly suited to economic, financial, and environmental datasets, where skewness and extreme values are common. By modeling heterogeneous impacts, MMQR offers detailed insights into the distributional characteristics of variables, such as risk, return, and policy effects that mean-based methods may overlook. The method also enables analysis of median and tail quantiles, facilitating the identification of possible opposing consequences or extreme risks. As highlighted by Li et al. (2025), MMQR employs a conditional quantile location-scale depiction and improving practical relevance. Formally, for a given set of covariates x_i , the conditional quantile of y_i can be expressed in Eq. (17) as follows:

$$Q_{y_i}(\tau/X_i) = X_i^T \beta_\tau \quad (17)$$

Nevertheless, traditional quantile regression does not address the issue of latent heterogeneity. It is observed that the unobserved heterogeneity can be incorporated using theoretically grounded extensions (Li et al., 2025). Accordingly, the quantile regression model is modified to account for unobserved heterogeneity reported in Eq. (18) as follows:

$$Q_{y_i}(\tau|\alpha_i, X_i) = \alpha_i + X_{it}'\beta(\tau_k) \quad (18)$$

In this study, the application of the MMQR is justified by its effectiveness in addressing key econometric challenges, including unobserved heterogeneity, and potential endogeneity issues that commonly arise in financial and economic data characterized by asymmetric relationships. In particular, the conditional quantiles of the explained time series, $Q_{y_i}(\tau/X_{it})$, are assessed across various distributional points by means of the advanced panel quantile regression approach proposed by Machado and Silva (2019). This methodology provides deeper insights into the underlying data structure, allowing for a comprehensive examination of distributional dynamics and heterogeneous effects within the panel framework. The formal specification of the model is presented in Eq. (19) as follows:

$$Q_{y_i}(\tau/X_{it}) = (\alpha_i + \delta_i\rho(\tau)) + X'_{it}\beta + Z'_{it}\gamma\rho(\tau) \quad (19)$$

Here, X_{it} denotes the set of explanatory variables, while $Q_{y_i}(\tau/X_{it})$ shows the conditional quantile of the dependent variable given X_{it} . The term $\alpha_i + \delta_i\rho(\tau)$ captures the fixed effect at quantile τ , $\rho(\tau)$ corresponding to the τ_{th} quantile, estimated via the following optimization procedure reported in Eq. (20) as follows:

$$\min_{(e_\rho)} \sum_i \sum_t \rho_T(R_{it} - (\delta_i + Z'_{it}\gamma)q) \quad (20)$$

The estimation function is formally expressed in Eq. (20) as follows:

$$\rho_T(R) = (T-1)RI\{R \leq 0\} + TRI\{R > 0\} \quad (21)$$

3.2.6 Panel second-generation causality test

After estimating the long-term coefficients, this study proceeds to evaluate the causality direction among the variables. It employs the panel non-causality framework developed by Dumitrescu and Hurlin (2012), which overcomes the CD and SH restrictions commonly linked with conventional causality tests (Granger, 1969). This approach develops a Wald test statistic, signified ($W_{N.T}^{HNC}$), which is functional transversely all cross-sectional components and series to evaluate causal links. The mathematical specification of the panel causality approach is presented in Eq. (22) as follows:

$$Y_{it} = \alpha_i + \sum_{j=1}^J \beta_i^j Y_{i(t-j)} + \sum_{j=1}^J \gamma_i^j X_{i(t-j)} + \mu_{it} \quad (22)$$

In this specification, (X) and (Y) represent the explanatory and explained variables, while γ_i^j and β_i^j correspond to the long-run parameters and AR coefficients. Both hypotheses (H_0 and H_1) of the D-H non-causality approach are formally stated in Eqs. (23) and (24), respectively.

$$\text{Null hypothesis} - H_0: \mu_i = 0 \quad \text{for } \forall i \quad (23)$$

$$\text{Alternative hypothesis} - H_1: \begin{cases} \mu_i = 0 & \text{for all } i = 1, 2, \dots, N_1 \\ \mu_i \neq 0 & \text{for all } i = N_1 + 1, 2, \dots, N \end{cases} \quad (24)$$

The Wald¹ test statistic is utilized to assess the hypothesis of Granger causality approach. Fig. 8 presents the methodological relationship schema of the study.

4. Estimated results and discussion

4.1 Preliminary tests (descriptive and correlation analysis)

¹ $W_{N.T}^{HNC} = N^{-1} \sum_{i=1}^N W_{i,T}$

Table 1 shows the descriptive statistics for all variables included in the study, covering 116 observations across the sample countries. The explained variable, LCF, which measures environmental sustainability, has an average of 0.699 and a median of 0.406, representing a right-skewed distribution with some countries exhibiting relatively high environmental capacity. This is supported by a skewness of 1.18 and a kurtosis of 2.80. Moreover, AI development exhibits a mean of 5978 and a median of 1495, highlighting substantial variation across countries, with the maximum reaching over 20,700 patents. The distribution is moderately right-skewed (0.58) and platykurtic (1.81), reflecting concentration in a few highly innovative countries. Additionally, NR show considerable skewness (2.06) and high kurtosis (8.18), indicating extreme values in some resource-dependent economies, while GTI displays wide dispersion with a mean of 18,057 and a median of 10,510 patents. The skewness of GTI is moderate (0.34), reflecting some asymmetry in innovation outcomes. Also, FD averages 0.755 with a slightly negative skewness (−0.79), suggesting a minor concentration in countries with above-average financial development. Furthermore, REC has a mean of 15.47%, with a pronounced right skew (1.22) due to high adoption levels in a subset of countries. Besides, Jarque-Bera tests confirm significant deviations from normality for all variables ($p < 0.01$), justifying the use of natural logarithmic transformations in subsequent analyses to reduce heteroscedasticity and stabilize variance. These descriptive statistics reveal substantial heterogeneity in technological, financial, and environmental indicators across the sampled countries, underscoring the need for econometric techniques that account for distributional and cross-sectional differences.

Table 2 reports the bivariate correlation among the key study variables, revealing important insights into the associations between financial, technological, and environmental indicators. The LCF is negatively correlated with both AI (−0.508) and GTI (−0.491), suggesting that higher levels of technological development and innovation are linked with reduced ecological pressure across the sampled countries. In contrast, LCF shows a positive correlation with NR (0.588) and FD (0.331), representing that resource utilization and a well-developed financial sector can enhance environmental capacity. In addition, AI is strongly positively correlated with GTI (0.628) and FD (0.459), reflecting the synergistic relationship between innovation, artificial intelligence development, and financial infrastructure. However, AI is negatively correlated with NR (−0.600) and REC (−0.509), highlighting potential trade-offs between resource dependence, renewable energy adoption, and technological advancement. Similarly, REC exhibits significant negative correlations with GTI (−0.600) and FD (−0.627), while showing a modest positive association with NR (0.209). These results underscore the complex and interdependent nature of environmental, technological, and financial factors, emphasizing the need for econometric methods that account for heterogeneity and CD when analyzing the determinants of environmental sustainability.

4.2 Results of cross-sectional dependency tests

Table 3 reports the results of several advanced CD tests, including the Pesaran CD, CDw, CDw+, and CD* statistics, applied to all study variables. These tests are designed to detect interdependencies across cross-sectional units and are particularly appropriate for panels characterized by heterogeneity and common shocks. The results from the Pesaran CD and CDw tests indicate statistically significant CD for most variables. Specifically, LLCF exhibits significant dependence at the 5% level, while LAI, LNR, LGTI, and LFD show strong dependence at the 1%

level. In contrast, LREC does not display significant dependence under the standard CD and CDw tests, suggesting weaker interconnections across countries for renewable energy when using these conventional measures. However, when applying the more powerful CDw+ and CD* tests, strong CD is detected for all variables, including LREC, at the 1% significance level. The large and highly significant CDw+ and CD* statistics indicate the presence of common factors, spillover effects, or shared shocks across countries, which are not fully apprehended by traditional CD tests. The results provide robust evidence of CD across the panel. This confirms that shocks affecting one country are expected to spill over to others, reflecting economic integration, technological diffusion, and environmental interlinkages among the sampled economies. These results strongly rationalize the use of second-generation panel econometric methods that obviously account for CD and avoid biased and inconsistent inference.

4.3 Results of slope heterogeneity tests

Table 4 shows the results of SH tests using the Pesaran and Yamagata (2008) and Blomquist and Westerlund (2013) approaches. The findings indicate significant heterogeneity across panel units for the study variables. Specifically, the Pesaran-Yamagata $\tilde{\Delta}$ statistic and its adjusted form (5.752 and 6.604) are both highly significant, suggesting substantial variation in slope coefficients across countries. Similarly, the Blomquist-Westerlund $\tilde{\Delta}$ statistics (2.358 and 2.707) confirm heterogeneity. These results highlight the need for econometric methods that account for varying slopes to avoid biased and inconsistent estimates in panel analyses.

4.4 Results of panel unit root tests

Table 5 presents the findings of the second-generation panel unit root approaches using the CADF and CIPS procedures, both of which explicitly account for CD across panel units. The tests are conducted under specifications with an intercept only and with both an intercept and a linear trend. The CADF findings show that, at levels, most variables are unit root under both specifications, as reflected by statistically insignificant test statistics. An exception is LFD, which appears weakly stationary at the 10% level under the intercept-only model and at the 5% level when a trend is involved. However, after first differencing, all series have become strongly stationary at the 1% level of significance across both model specifications. This confirms that the variables are integrated of order one $I(1)$. The CIPS test results provide consistent evidence. At levels, most series fail to reject the null hypothesis of a unit root, except for LGTI and LFD, which exhibit stationarity under certain specifications. In contrast, all variables become highly significant at first differences under both intercept and intercept–trend at the 1% level. The combined CADF and CIPS findings robustly suggest that the variables are predominantly $I(1)$. This result justifies the application of second-generation panel cointegration techniques, such as the Westerlund test, to scrutinize long-run stable equilibrium association amongst the selected variables while accounting for CD and slope heterogeneity.

4.5 Results of cointegration tests

Table 6 reports the results of panel long-run cointegration tests using the Kao, Pedroni, and Westerlund approaches. The Kao test strongly rejects the null of no cointegration across all

statistics, including the Modified Dickey–Fuller t ($-5.050, p < 0.01$) and the Augmented Dickey–Fuller t ($-4.107, p < 0.01$), indicating robust long-run associations among the variables in the full panel. Similarly, the Pedroni test largely confirms cointegration, with the Phillips–Perron t ($-9.187, p < 0.01$) and Augmented Dickey–Fuller t ($-2.831, p < 0.01$) demonstrating significant long-run links, while the Modified Phillips–Perron t ($1.556, p = 0.0599$) is slightly 10% significance level. In contrast, the Westerlund (2005) test detects cointegration only in a subset of panels (variance ratio = $-1.419, p = 0.078$), highlighting cross-sectional heterogeneity and differences in adjustment dynamics across countries. These results suggest that while the panel as a whole exhibits stable long-run relationships among technological, financial, resource, and environmental indicators, the significance and strength of cointegration vary across units. This underscores the importance of using econometric methods that account for heterogeneity and CD when analyzing multi-country panel data.

4.6 Results of long-run elasticity estimates

The MMQR results presented in Table 7 presents a nuanced understanding of how LAI, LNR, LGTI, LFD, and LREC influence the LCF indicator. By estimating coefficients across conditional quantiles of the dependent variable, the MMQR method captures the heterogeneous impacts of these variables, revealing relationships that standard mean-based regressions may overlook. The analysis demonstrates pronounced asymmetry in the effects of the independent variables, highlighting the importance of country-specific factors, distributional dynamics, and environmental capacity in shaping the effectiveness of technological, financial, and policy interventions.

The results show that LAI consistently exerts an adverse influence on LCF across all quantiles, with coefficients ranging from -0.240 at the 10th quantile to -0.305 at the 90th quantile. This adverse coefficient indicate that the expansion of AI development is linked with a decline in the LCF, suggesting that increased AI activity may exert additional pressure on ecological sustainability. Moreover, the magnitude of this negative relationship becomes more distinct at higher quantiles of LCF, implying that nations with comparatively stronger environmental performance may experience greater environmental strain as AI adoption intensifies. These results are reliable with earlier empirical studies that highlight the substantial environmental costs associated with AI development (Strubell et al., 2019; Luan et al., 2022; Cowls et al., 2023; Alnafrh, 2025). One plausible explanation for this relationship lies in the highly energy-intensive nature of AI technologies, which rely heavily on cloud computing systems, large-scale data centers, and high-performance computing infrastructures. The operation, training, and deployment of advanced machine learning models demand substantial computational resources, leading to substantial electricity use and, in many cases, increased carbon emissions, particularly in regions where energy systems remain dependent on fossil fuels. Although AI technologies hold considerable potential to improve environmental outcomes through applications such as optimizing resource allocation, enhancing energy efficiency, enabling smart grid management, and supporting real-time environmental monitoring, the immediate environmental footprint associated with AI-driven digital and industrial expansion may outweigh these benefits in the short run. This effect becomes especially evident in economies with higher levels of technological advancement and digital infrastructure, where AI deployment is more widespread and computational demand is

significantly greater. Therefore, the growing magnitude of the negative effect across higher quantiles recommends that countries with relatively strong environmental performance may paradoxically face greater ecological pressures as AI adoption accelerates. These results highlight the quick need for representatives to integrate AI development within sustainable energy frameworks, encourage the consumption of renewable energy sources to implement robust regulatory mechanisms and power digital infrastructure aimed at minimizing the environmental externalities of AI technologies. Such measures are vital to confirm that the long-term benefits of AI innovation contribute positively to environmental sustainability rather than exacerbating environmental degradation.

In contrast, LNR demonstrate a consistently positive effect on environmental quality, with coefficients ranging from 0.305 at the 10th quantile to 0.330 at the 90th quantile. These results are in line with previous empirical evidence (Guo et al., 2024; Sun et al., 2024). This pattern indicates that higher resource rents contribute to enhancing load capacity, likely through investments in environmental protection, sustainable infrastructure, and clean technologies. The slight increase in the effect size toward higher quantiles suggests that countries with better environmental outcomes are able to more efficiently allocate resource revenues to promote ecological sustainability. These findings challenge the conventional “resource curse” narrative by highlighting the potential for resource rents to support environmental management, particularly when institutional quality, governance frameworks, and strategic reinvestment policies are in place. Furthermore, the stability of the coefficients across quantiles indicates that the positive environmental contributions of natural resources are significant across nations with varying levels of environmental quality. This reinforces the view that resource wealth can be strategically leveraged to promote ecological improvement when managed prudently.

Similarly, LGTI displays a positive and progressively increasing impact on LCF across the conditional distribution, with coefficients moving from 0.093 at the 10th quantile to 0.249 at the 90th quantile. The smaller effects observed at lower quantiles likely reflect structural limitations in countries with weaker environmental performance, such as insufficient absorptive capacity, limited infrastructure, or weaker policy support for green technologies. These findings align with previous studies (Degirmenci and Aydin, 2024; Sun et al., 2024; Uche et al., 2024). As countries move to higher quantiles, the effectiveness of green innovation strengthens significantly, suggesting that these economies can better influence technological advancements to improve energy efficiency, reduce emissions, and implement cleaner production processes. This distributional pattern underscores the importance of supporting not only the development of green technologies but also the enabling environments, such as institutional quality, regulatory frameworks, and knowledge networks that allow these innovations to translate into tangible environmental benefits. By highlighting the heterogeneity in LGTI impacts, the results suggest that policy interventions should be tailored to national environmental and technological capacities to maximize ecological gains.

The role of LFD emerges as particularly critical, with coefficients showing the strongest positive impact among all variables. The location estimate indicates a coefficient of 1.206, while quantile-specific estimates rise from 1.033 at q10 to 1.406 at q90. These results are consistent with previous studies (Zafar et al., 2022; Adebayo et al., 2025; Dar et al., 2025). These results indicate that

financial systems are key facilitators of environmental sustainability, providing the capital necessary for investments in alternative and renewable energy, green infrastructure, and eco-friendly technologies. The increasing magnitude of the effect at higher quantiles suggests that countries with stronger environmental performance are better able to channel financial resources toward sustainable initiatives, reflecting a complementary relationship between financial development and ecological capacity. A deeper and more efficient financial markets enable governments and private actors to mobilize long-term financing for environmental projects, reduce transaction costs, and encourage innovation adoption. These results highlight the essential for policy frameworks that support financial institutions, enhance access to green finance, and promote the alignment of investment incentives with environmental objectives.

Ina addition, LREC also shows a positive and increasingly strong effect on LCF across quantiles. Coefficients for LREC range from 0.421 at q10 to 0.764 at q90, highlighting the scale-dependent benefits of renewable energy integration. At lower quantiles, where countries exhibit relatively lower environmental performance, the impact of renewable energy is moderate, potentially reflecting limited infrastructure, lower penetration of green technologies, or continued dependance on non-renewables. In contrast, at higher quantiles, the substantial positive effect suggests that countries with greater environmental capacity are better positioned to leverage renewable energy to diminish emissions, boost energy efficiency, and enhance ecological resilience. These findings are aligned with the earlier findings (Sun et al., 2024; Chen et al., 2025; Yu et al., 2025; Li et al., 2025). The rising magnitude of LREC coefficients reinforces the idea that renewable energy adoption yields increasing returns as a country's environmental and technological capabilities improve, emphasizing the need for supportive policies that accelerate clean energy transitions across all levels of environmental performance.

Fig. 9 presents the MMQR results and demonstrates that the effects of the explanatory variables vary across different quantiles of the LCF in Quad economies. The AI exhibits a negative impact across all quantiles, suggesting that increased AI activities may exert environmental pressure. In contrast, NR, GTI, FD, and REC positively influence LCF, with their effects generally strengthening at higher quantiles. These findings indicate that natural resources, green technological innovations, financial development, and renewable energy play important roles in enhancing environmental sustainability, while the environmental implications of AI require careful management in the Quad economies.

These analyses underscore that the drivers of environmental sustainability are heterogeneous and distribution-dependent. AI poses consistent challenges, whereas green innovation, natural resources, financial expansion, and renewable energy collectively enhance environmental outcomes, with their effects increasing for countries with higher ecological capacity. By providing quantile-specific insights, these results allow policymakers to design targeted, differentiated, and context-sensitive strategies to balance technological advancement, economic development, and environmental sustainability across countries and regions.

4.7 Results of Dumitrescu Hurlin causality test

Table 8 displays the findings of the Dumitrescu–Hurlin (D-H) panel causality tests with one lag, examining the causality direction among the main variables such as LLCF, LAI, LNR, LGTI, LFD,

and LREC. The findings reveal multiple significant causal relationships, highlighting the dynamic interplay between technological, financial, and environmental factors. The findings show bidirectional causality between LAI and LLCF. The LAI significantly Granger-causes LLCF at the 1% level (W-Stat = 3.437), while LLCF also influences LAI (W-Stat = 5.261), suggesting that advancements in AI both affect environmental sustainability and are shaped by ecological capacity. On one hand, AI deployment influences LCF through both direct and indirect environmental pathways. AI-driven technologies, such as high-performance computing, cloud infrastructure, and automation systems consume substantial energy, which, if sourced from fossil fuels, upsurges carbon emissions and reduces ecological capacity. Simultaneously, AI can enhance environmental performance by optimizing industrial processes, enlightening the efficiency of energy sources, supporting alternative and renewable energy integration, and enabling advanced monitoring of biological systems. On the other hand, the state of environmental quality, captured by LCF, can shape AI adoption and deployment strategies. The Quad economies reflecting greater ecological resilience may attract increased investment in AI technologies due to the availability of sustainable energy and regulatory frameworks encouraging to low-carbon innovation. Conversely, countries facing environmental constraints may adjust AI implementation, prioritizing energy-efficient algorithms, green data centers, or AI applications in resource management.

Furthermore, the results indicate that LNR exert a unidirectional causal influence on the LLCF (W-Stat = 5.157), whereas LLCF does not have a substantial effect on LNR. This finding recommends that the exploitation and management of natural resources play a decisive role in determining environmental consequences, while environmental capacity, in turn, does not significantly alter the generation or allocation of resource rents. Efficiently, resource rents act as a driver of ecological performance rather than responding to changes in environmental sustainability. Similarly, LFD and LREC significantly Granger-cause LLCF, while LCF only weakly influences LFD and not LREC, highlighting the role of financial development and renewable energy adoption in enhancing environmental sustainability. For LAI, LNR, LGTI, LFD, and LREC, the tests reveal significant causal links in several cases. Notably, LREC shows strong causality toward LAI, LNR, and LGTI, reflecting the position of renewable energy adoption in promoting technological modernization and resource management. Likewise, LAI is also influenced by LGTI, LFD, and LREC, emphasizing the feedback effects between innovation, financial development, and sustainable energy. The DH causality analysis underscores the interconnectedness of technology, finance, natural resources, and energy in shaping environmental outcomes. The existence of both bidirectional and unidirectional causality underscores the importance of adopting integrated policy measures that simultaneously promote artificial intelligence, green innovation, financial progress, and renewable energy to achieve sustainable ecological performance.

5. Conclusion and policy suggestions

This study scrutinizes the complex relationships between LAI adoption, LNR, LGTI, LFD, LREC, and the LCF across the Quad economies over the period 1995–2023. After detecting the significant cross-sectional dependency, thus study applied second-generation econometric process. Moreover, to account for heterogeneous impacts across the distribution of ecological outcomes, the analysis utilizes the advanced MMQR framework. The results reveal that AI adoption exerts a consistently

adverse effect on environmental sustainability, as measured by the load capacity factor, with the adverse impact intensifying at higher quantiles. In contrast, green technological innovation, natural resource rents, financial development, and renewable energy positively influence LCF across all quantiles, with effects generally strengthening for countries exhibiting higher environmental performance. These results highlight significant distributional heterogeneity, indicating that the magnitude of each factor's impact depends on the country's position within the ecological spectrum. The findings suggest that while AI presents environmental challenges, strategic investments in green innovation, renewable energy, and financial development can mitigate its negative effects and enhance environmental capacity. In addition, the pairwise DH causality results, revealing significant dynamic interactions among the variables. Particularly, AI and LCF exhibit bidirectional causality, indicating mutual influence between technological advancement and environmental capacity. The LNR, LFD, and LREC show unidirectional causality toward LCF, highlighting their role in enhancing sustainability. Moreover, LREC also drives LAI, LNR, and LGTI, while LAI is shaped by LGTI, LFD, and LREC, emphasizing feedback effects. Furthermore, natural resource revenues, when managed effectively, contribute positively to sustainability outcomes, highlighting the importance of integrating technological, financial, and policy interventions to achieve balanced economic and ecological growth across the Quad economies.

The MMQR findings provide critical insights for policy formulation across the Quad economies, highlighting the need for tailored strategies that balance technological advancement, financial development, resource management, and environmental sustainability. For Australia, a country with abundant natural resources and a strong renewable energy sector, policy implications should prioritize leveraging resource rents and renewable energy expansion to strengthen ecological capacity. The natural resource revenues and renewable energy adoption substantially enhance load capacity. Policymakers should channel mining and energy rents into sustainable infrastructure, carbon-neutral technologies, and ecological restoration projects. Given the observed negative impact of AI on LCF, it is vital for Australia to integrate AI progress with renewable energy deployment and strict environmental regulations. Investments in green AI, energy-efficient data centers, and digital infrastructures powered by renewable energy could mitigate potential ecological strain while enabling technological growth. Moreover, supporting green technological innovation through research and development (R&D) incentives and public-private partnerships can enhance environmental outcomes, particularly in sectors such as smart grids, precision agriculture, and low-emission manufacturing.

In India, which faces substantial environmental pressures from population growth and rapid industrialization, the findings emphasize the critical need to simultaneously advance green innovation, financial expansion, and renewable energy to strengthen long-term ecological capacity. The empirical outcomes indicate that green technologies and financial sector development have strong positive effects on environmental outcomes, highlighting the potential for India to enhance sustainability by reinforcing the financial system to support green investments. Policy measures should prioritize expanding access to green financing for renewable energy projects, energy-efficient infrastructure, and pollution control technologies. In light of AI's potential environmental impact, India should implement regulatory frameworks that prevent AI deployment from exacerbating energy consumption or emissions. At the same time, targeted

initiatives, such as incentivizing AI applications in resource optimization, smart energy management, and low-carbon technologies can mitigate adverse effects while leveraging AI to boost productivity. Furthermore, directing natural resource rents toward sustainable development programs can provide essential funding for environmental conservation, renewable energy expansion, and climate adaptation efforts. By integrating these strategies, India can promote a balanced approach that harnesses technological innovation while safeguarding ecological resilience.

In Japan, where technological advancement and financial sophistication are among the highest globally, the policy focus should be on integrating AI, green innovation, and renewable energy within a highly developed institutional framework. The adverse influence of AI on environmental quality indicates the need to ensure that AI deployment, particularly in industrial automation, robotics, and big data analytics is powered by renewable energy and aligned with strict environmental standards. Japan's strong innovation ecosystem offers an opportunity to leverage green technological innovation to improve energy efficacy and diminish emissions in manufacturing and transport sectors. Financial development can be further leveraged to facilitate large-scale investments in clean energy infrastructure, carbon capture, and sustainable urban planning. Moreover, although Japan has limited natural resource rents compared to other Quad economies, its strategic use of resource imports and recycling initiatives can supplement sustainability efforts, ensuring that technological and energy investments deliver maximum environmental benefits. Accelerating renewable energy adoption, particularly offshore wind, solar, and hydrogen technologies, will be essential to offset the ecological pressures associated with AI and industrial activity.

In the United States, a global leader in technological innovation and financial markets, evidence indicates that AI deployment presents notable environmental risks. However, these risks can be effectively mitigated through integration with renewable energy sources and the implementation of robust environmental regulations. Policymakers should prioritize the decarbonization of AI-intensive sectors, such as cloud computing, data analytics, and automation by aligning AI development with low-carbon energy solutions, including solar and wind power. Although natural resource rents are relatively moderate, they can be strategically leveraged to support environmental conservation, accelerate the energy transition, and encourage research and development in clean technologies. The observed positive influences of renewable energy and financial development suggest that expanding access to green finance and incentivizing private-sector investment in green technologies can further strengthen ecological outcomes. Moreover, the United States should continue fostering green technological innovation in areas such as energy efficiency, smart grid systems, and electric vehicles. Ensuring that these innovations complement AI deployment and resource management strategies will be critical to achieving sustainable environmental outcomes.

The Quad economies face both opportunities and challenges in balancing AI adoption, natural resource management, technological innovation, financial development, and renewable energy integration. Country-specific policies should emphasize the mitigation of AI's negative ecological impacts, promotion of green technologies, efficient utilization of resource rents, strengthening of financial systems for sustainable investment, and accelerated deployment of renewable energy. Australia and India should focus on leveraging natural resources and renewable energy expansion,

with India also emphasizing green financing and institutional support. Japan should integrate AI deployment with green innovation and renewable energy within a highly developed technological and regulatory environment. The United States should focus on decarbonizing AI-intensive sectors and promoting green technological solutions, supported by strong financial mechanisms and policy incentives. These tailored interventions can help the Quad economies achieve sustainable development goals, improve load capacity, and balance technological and economic growth with ecological preservation.

While the present study offers valuable insights into the AI–resource–environment nexus in Quad economies, numerous limitations highlight avenues for future research. First, although the MMQR framework captures distributional heterogeneity, the analysis is predominantly linear. Future work could explore non-linearities and threshold effects to identify critical levels of AI adoption, resource rents, financial development, renewable energy, or green technological innovation that alter the magnitude or direction of environmental impacts, revealing potential tipping points in sustainable environment transitions. Second, the study focuses on core variables (AI, natural resource rents, financial development, renewable energy, and green technologies), while excluding institutional, economic, and policy dimensions. Incorporating governance quality, regulatory stringency, human capital, green finance, and innovation systems could deliver a more inclusive understanding of the conditions that enable effective environmental sustainability and may interact with the examined factors. Third, the aggregate analysis of Quad economies captures broad trends but may mask sectoral or regional heterogeneity. Future research could extend the framework to other country groups or sector-specific contexts to uncover context-dependent patterns. Finally, complementary methodologies, such as spatial econometrics, structural models, or firm-level analyses of AI, green innovation and resource use could further elucidate causal mechanisms and strengthen policy-relevant insights.

References

- Adebayo, T.S., Eweade, B.S., Özkan, O., Uzun Ozsahin, D., 2025. Effects of energy security and financial development on load capacity factor in the USA: A wavelet kernel-based regularized least squares approach. *Clean Technol. Environ. Policy* 27 (9), 4215-4232. <https://doi.org/10.1007/s10098-024-03109-1>.
- Ahmad, M., Jiang, P., Majeed, A., Umar, M., Khan, Z., Muhammad, S., 2020. The dynamic impact of natural resources, technological innovations and economic growth on ecological footprint: an advanced panel data estimation. *Resour. Policy* 69, 101817. <https://doi.org/10.1016/j.resourpol.2020.101817>.
- Ahmed, Z., Pinzon, S., Rasheed, M.Q., 2025. Unlocking marginal SDG dynamics: a KRLS machine learning analysis of AI technologies and solar energy. *Geosci. Front.* 102167. <https://doi.org/10.1016/j.gsf.2025.102167>.

- Alnafrah, I., 2025. The Two Tales of AI: A Global assessment of the environmental impacts of artificial intelligence from a multidimensional policy perspective. *J. Environ. Manage.* 392, 126813. <https://doi.org/10.1016/j.jenvman.2025.126813>.
- Blomquist, J., Westerlund, J., 2013. Testing slope homogeneity in large panels with serial correlation. *Econ. Lett.* 121 (3), 374–378. <https://doi.org/10.1016/J.ECONLET.2013.09.012>.
- Chen, W., Usman, M., Kousar, R., Ahmad, P., 2025. How digitalization, renewable energy, and natural resources shape the environmental excellence? Evidence from China using a Quantile-on-Quantile framework. *Geosci Front.* 102055. <https://doi.org/10.1016/j.gsf.2025.102055>.
- Cowls, J., Tsamados, A., Taddeo, M., Floridi, L., 2023. The AI gambit: leveraging artificial intelligence to combat climate change—opportunities, challenges, and recommendations. *AI Soc.* 38(1), 283-307. <https://doi.org/10.1007/s00146-021-01294-x>.
- Dar, Y.A., Malik, M.H., Rasool, H., 2025. Exploring the Load Capacity Factor and Pollution Haven Hypothesis in India: Analyzing Energy Security Risk and Financial Development in an Asymmetric Framework. *Environ. Qual. Manage.* 34(4), e70091. <https://doi.org/10.1002/tqem.70091>.
- Degirmenci, T., Aydin, M., 2024. Testing the load capacity curve hypothesis with green innovation, green tax, green energy, and technological diffusion: A novel approach to Kyoto protocol. *Sustain. Dev.* 32(5), 4931-4945. <https://doi.org/10.1002/sd.2946>.
- Dumitrescu, E.I., Hurlin, C., 2012. Testing for Granger non-causality in heterogeneous panels. *Econ. Model.* 29 (4), 1450-1460. <https://doi.org/10.1016/j.econmod.2012.02.014>.
- Fan, J., Li, W., Chen, L., 2025. How does artificial intelligence affect energy efficiency? Evidence from supply chain digitization pilot program. *Energy Econ.* 108728. <https://doi.org/10.1016/j.eneco.2025.108728>.
- Fan, J., Liao, Y., Yao, J., 2015. Power enhancement in high-dimensional cross-sectional tests. *Econometrica* 83 (4), 1497-1541. <https://doi.org/10.3982/ECTA12749>.
- Galli, A., 2015. On the rationale and policy usefulness of Ecological Footprint Accounting: The case of Morocco. *Environ. Sci. Policy* 48, 210-224. <https://doi.org/10.1016/j.envsci.2015.01.008>.
- GFPN., 2024. Global footprint network database. <https://www.footprintnetwork.org/our-work/ecological-footprint>. (Accessed on 29 June 2025).
- Granger, C.W., 1969. Investigating causal relations by econometric models and cross-spectral methods. *Econometrica: J. Econometric Soc.* 37 (3), 424-438. <https://doi.org/10.2307/1912791>.
- Guo, Y., Alnafrah, I., Hou, X., A. Almenweer, R., Mehmood, U., 2025. How natural resources and technological advancements affect environmental quality: Evidence from load capacity

- factor in N-11 nations. *Nat. Resour. Forum* 49:2, 1769-1794. <https://doi.org/10.1111/1477-8947.12442>.
- Haseeb, M., Usman, M., Fan, F., 2025. Understanding financial stability through the lens of geopolitical risk in China: The influence of policy uncertainty on climate change. *Finance Res. Lett.* 107944. <https://doi.org/10.1016/j.frl.2025.107944>.
- Hussain, M., Dogan, E., 2021. The role of institutional quality and environment-related technologies in environmental degradation for BRICS. *J. Clean. Prod.* 304, 127059. <https://doi.org/10.1016/j.jclepro.2021.127059>.
- IMF., 2024. International Monetary Fund (IMF) database. [https://data.imf.org/en/Data-Explorer?datasetUrn=IMF.MCM:FDI\(1.0.0\)](https://data.imf.org/en/Data-Explorer?datasetUrn=IMF.MCM:FDI(1.0.0)). (Accessed on 29 June 2025).
- IPCC., 2018. Summary for Policymakers. In: *Global Warming of 1.5°C*. <https://www.ipcc.ch/sr15/chapter/spm>. (Accessed on 22 September 2025).
- Iqbal, W., Tang, Y.M., Lijun, M., Chau, K.Y., Xuan, W., Fatima, A., 2021. Energy policy paradox on environmental performance: the moderating role of renewable energy patents. *J. Environ. Manage.* 297, 113230. <https://doi.org/10.1016/j.jenvman.2021.113230>.
- Juodis, A., Reese, S., 2022. The incidental parameters problem in testing for remaining cross-section correlation. *J. Bus. Econ. Stat.* 40 (3), 1191-1203. <https://doi.org/10.1080/07350015.2021.1906687>.
- Kao, C., 1999. Spurious regression and residual-based tests for cointegration in panel data. *J. Econom.* 90 (1), 1-44. [https://doi.org/10.1016/S0304-4076\(98\)00023-2](https://doi.org/10.1016/S0304-4076(98)00023-2).
- Kartal, M.T., 2024. Impact of environmental tax on ensuring environmental quality: Quantile-based evidence from G7 countries. *J. Clean. Prod.* 140874. <https://doi.org/10.1016/j.jclepro.2024.140874>.
- Kartal, M.T., Mukhtarov, S., Kirikkaleli, D., 2025. Achieving environmental quality through stringent environmental policies: Comparative evidence from G7 countries by multiple environmental indicators. *Geosci Front.* 16 (1), 101956. <https://doi.org/10.1016/j.gsf.2024.101956>.
- Kartal, M.T., Shahbaz, M., Taşkın, D., Kılıç Depren, S., Ayhan, F., 2024b. How are energy transition and energy-related R&D investments effective in enabling decarbonization? Evidence from Nordic Countries by novel WLMC model. *J. Environ. Manage.* 365, 121664. <https://doi.org/10.1016/j.jenvman.2024.121664>.
- Kartal, M.T., Taşkın, D., Shahbaz, M., Kirikkaleli, D., Kılıç Depren, S., 2024a. Role of energy transition in easing energy security risk and decreasing CO2 emissions: disaggregated level evidence from the USA by quantile-based models. *J. Environ. Manage.* 359, 120971. <https://doi.org/10.1016/j.jenvman.2024.120971>.

- Li, Y., Alharthi, M., Ahmad, I., Hanif, I., Hassan, M.U., 2022. Nexus between renewable energy, natural resources and carbon emissions under the shadow of transboundary trade relationship from South East Asian economies. *Energy Strat. Rev.* 41, 100855. <https://doi.org/10.1016/j.esr.2022.100855>.
- Li, Z., Usman, M., Makhdum, M.S.A., Alomair, A., 2025. A double-edged sword? Exploring the heterogeneous effects of industrialization, energy efficiency, and disaggregated financial development on environmental management. *J. Environ. Manage.* 393, 126926. <https://doi.org/10.1016/j.jenvman.2025.126926>.
- Luan, F., Yang, X., Chen, Y., Regis, P.J., 2022. Industrial robots and air environment: A moderated mediation model of population density and energy consumption. *Sustain. Prod. Consum.* 30, 870-888. <https://doi.org/10.1016/j.spc.2022.01.015>.
- Machado, J.A., Silva, J.S., 2019. Quantiles via moments. *J. Econom.* 213 (1), 145-173. <https://doi.org/10.1016/j.jeconom.2019.04.009>.
- Mahajan, M., Sah, A.N., 2026. Managing emissions and mitigating global warming through collaborative emission control: Empirical evidence from QUAD countries. *Sustain. Dev.* 1-32. <https://doi.org/10.1002/sd.70878>.
- Niu, X., Lin, C., He, S., Yang, Y., (2025). Artificial intelligence and enterprise pollution emissions: From the perspective of energy transition. *Energy Econ.* 144, 108349. <https://doi.org/10.1016/j.eneco.2025.108349>.
- OECD., 2024. Data of Patents on Environment Technologies (Indicator). <https://www.oecd.org/en/data/indicators/patents-on-environment-technologies.html>. (Accessed on 29 June 2025).
- Ozcan, B., Kılıç Depren, S., Kartal, M.T., 2023. Impact of nuclear energy and hydro electricity consumption in achieving environmental quality: Evidence from load capacity factor by quantile based non-linear approaches. *Gondwana Res.* <https://doi.org/10.1016/j.gr.2023.05.014>.
- Pedroni, P., 1999. Critical values for cointegration tests in heterogeneous panels with multiple regressors. *Oxf. Bull. Econ. Stat.* 61 (S1), 653-670. <https://doi.org/10.1111/1468-0084.0610s1653>.
- Pedroni, P., 2004. Panel cointegration: asymptotic and finite sample properties of pooled time series tests with an application to the PPP hypothesis. *Econom. Theory* 20 (3), 597-625. <https://doi.org/10.1017/S0266466604203073>.
- Pesaran, M.H., 2007. A simple panel unit root test in the presence of cross-section dependence. *J. Appl. Econom.* 22(2), 265-312. <https://doi.org/10.1002/jae.951>.

- Pesaran, M.H., 2015. Testing weak cross-sectional dependence in large panels. *Econom. Rev.* 34 (6-10), 1089-1117. <https://doi.org/10.1080/07474938.2014.956623>.
- Pesaran, M.H., 2021. General diagnostic tests for cross-sectional dependence in panels. *Empir. Econ.* 60 (1), 13-50. <https://doi.org/10.1007/s00181-020-01875-7>.
- Pesaran, M.H., Xie, Y., 2021. A Bias-Corrected CD Test for Error Cross-Sectional Dependence in Panel Data Models with Latent Factors. *Cambridge Working Papers in Econ.* 2158. <https://doi.org/10.48550/arXiv.2109.00408>.
- Pesaran, M.H., Yamagata, T., 2008. Testing slope homogeneity in large panels. *J. Econom.* 142(1), 50-93. <https://doi.org/10.1016/j.jeconom.2007.05.010>.
- Quad Leaders' Joint Statement., 2023. "Quad Leaders' Joint Statement." https://www.mea.gov.in/bilateral-documents.htm?dtl/36571/Quad_Leaders_Joint_Statement.
- Rees, W.E., 1992. Ecological footprints and appropriated carrying capacity: what urban economics leaves out. *Environ. Urban* 4 (2), 121–130. <https://doi.org/10.1177/095624789200400212>.
- Siche, R., Pereira, L., Agostinho, F., Ortega, E., 2010. Convergence of ecological footprint and emergy analysis as a sustainability indicator of countries: Peru as case study. *Commun. Nonlinear Sci. Numer. Simulat.* 15 (10), 3182-3192. <https://doi.org/10.1016/j.cnsns.2009.10.027>.
- Strubell, E., Ganesh, A., McCallum, A., 2019. Energy and policy considerations for deep learning in NLP. *Proc. 57th Annu. Meet Assoc. Comput. Linguist.* 3645-3650. <https://doi.org/10.48550/arXiv.1906.02243>.
- Sun, Y., Usman, M., Radulescu, M., Pata, U.K., Balsalobre-Lorente, D., 2024. New insights from the STIPART model on how environmental-related technologies, natural resources and the use of the renewable energy influence load capacity factor. *Gondwana Res.* 129, 398-411. <https://doi.org/10.1016/j.gr.2023.05.018>.
- Uche, E., Ngepah, N., Das, N., Dey, L., 2024. Dynamic interactions of green innovations, green transitions and ecological load capacity factor in BRICS. *Renew. Energy* 231, 120905. <https://doi.org/10.1016/j.renene.2024.120905>.
- Usman, M., Horobet, A., Radulescu, M., Balsalobre-Lorente, D., 2025. Environmental taxes, environmental policy stringency and policy complementarity: A comprehensive analysis of EU economic and environmental goals. *Int. Rev. Econ. Finance* 103, 104358. <https://doi.org/10.1016/j.iref.2025.104358>.
- Wang, X., Khan, M.U., Mir, F.W., 2025. Artificial intelligence as heterogeneous agents in environmental performance: The mediating role of energy transition and the moderating effect of green technology adoption in belt and road initiative countries. *J. Clean. Prod.* 523, 146331. <https://doi.org/10.1016/j.jclepro.2025.146331>.

- Wang, Y., Yuan, G., 2026. Natural resource exploitation and productive capacity as drivers of ecological footprint: The roles of technology and economic policy uncertainty. *Geosci. Front.* 102247. <https://doi.org/10.1016/j.gsf.2026.102247>.
- WDI from World Bank., 2024. World development indicators, World Bank Database. <https://databank.worldbank.org/source/world-development-indicators>. (Accessed on 29 June 2025).
- Westerlund, J., 2005. New simple tests for panel cointegration. *Econom. Rev.* 24 (3), 297-316. <https://doi.org/10.1080/07474930500243019>.
- WIPO., 2024. World Intellectual Property Organization (WIPO) Statistics Data Center. <https://www.wipo.int/portal/en/>. (Accessed on 29 June 2025).
- Xiaoman, W., Majeed, A., Vasbieva, D.G., Yameogo, C.E.W., Hussain, N., 2021. Natural resources abundance, economic globalization, and carbon emissions: Advancing sustainable development agenda. *Sustain. Dev.* 29 (5), 1037-1048. <https://doi.org/10.1002/sd.2192>.
- Yu, P., Abusamra, O.A., Usman, M., Makhdam, M.S.A., 2025. Leveraging technological innovation, renewable energy, and forest management to overcome barriers and achieve sustainable development Goal 13 (Climate Action) in G-7 countries. *Sustain. Dev.* 33 (6), 8821-8839. <https://doi.org/10.1002/sd.70127>.
- Zafar, M.W., Zaidi, S. A.H., Mansoor, S., Sinha, A., Qin, Q., 2022. ICT and education as determinants of environmental quality: The role of financial development in selected Asian countries. *Technol. Forecast. Soc. Chang.* 177, 121547. <https://doi.org/10.1016/j.techfore.2022.121547>.

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Table 1. Descriptive statistics of load capacity factor, artificial intelligence, natural resource rents, and control variables across the Quad countries.

	LCF	AI	NR	GTI	FD	REC
Mean	0.698952	5977.621	2.319629	18057.02	0.754819	15.46724
Median	0.406479	1495.000	1.763015	10510.00	0.860000	8.550000
Maximum	2.502915	20739.00	13.813495	2480.00	0.976000	49.50000
Minimum	0.112808	1.000000	0.012489	24.00000	0.367000	3.500000
Std. Dev.	0.695480	6713.195	2.737845	17283.80	0.202327	14.74037
Skewness	1.180232	0.580110	2.060125	0.336947	-0.788627	1.221521
Kurtosis	2.799952	1.809573	8.176974	1.422376	1.944680	2.807933
Jarque-Bera	27.12375	13.35560	211.5914	14.22465	17.40693	29.02582
Probability	0.000001	0.001259	0.000000	0.000815	0.000166	0.000000
Sum	81.07847	693404.0	269.0769	2094614	87.55900	1794.200
Sum Sq. Dev.	55.62470	5.18E+09	862.0163	3.44E+10	4.707673	24987.02
Observations	116	116	116	116	116	116

Note: Mean = average value; Std. Dev. = standard deviation; Sum Sq. Dev. = sum of square of standard deviation; LCF = Load Capacity Factor; AI = Artificial Intelligence; NR = Natural Resources; GTI = Green Technological Innovation; FD = Financial Development; REC = Renewable Energy Consumption.

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Table 2. Correlation matrix illustrating the pairwise associations among the selected variables, highlighting the strength and direction of their relationships.

Series	LCF	AI	NR	GTI	FD	REC
LCF	1.0000					
	–					
	–					
AI	-0.5082	1.0000				
	[-6.3006]	–				
	(0.0000)	–				
NR	0.5884	-0.6001	1.0000			
	[7.7709]	[-8.0113]	–			
	(0.0000)	(0.0000)	–			
GTI	-0.4912	0.6284	-0.6125	1.0000		
	[-6.0215]	[8.3897]	[-8.2753]	–		
	(0.0000)	(0.0000)	(0.0000)	–		
FD	0.3311	0.4588	0.0285	0.5021	1.0000	

[3.7467] [5.5145] [0.3048] [6.1985] –

(0.0003) (0.0000) (0.7610) (0.0000) –

REC -0.1827 -0.5089 0.2092 -0.5997 -0.6266 1.0000

[-1.9845][-6.3125] [2.2845] [-8.0016][-8.3548] –

(0.0496) (0.0000) (0.0242) (0.0000) (0.0000) –

Note: t-statistics and p-values are reported in square brackets [] and parentheses () respectively. Abbreviations: LCF = Load Capacity Factor; AI = Artificial Intelligence; NR = Natural Resources; GTI = Green Technological Innovation; FD = Financial Development; REC = Renewable Energy Consumption.

Table 3. Results of cross-sectional dependence tests showing the presence of interlinkages and interdependencies among the Quad countries.

	CD	CDw	CDw+	CD*
LLCF	-2.210** (0.027)	-2.210** (0.027)	20.640*** (0.000)	7.30e+30*** (0.000)
LAI	9.310*** (0.000)	9.310*** (0.000)	32.130*** (0.000)	2.60e+31*** (0.000)
LNR	4.110*** (0.000)	4.110*** (0.000)	17.950*** (0.000)	3.30e+30*** (0.000)
LGTI	3.710*** (0.000)	3.710*** (0.000)	14.660*** (0.000)	5.80e+30*** (0.000)
LFD	9.430*** (0.000)	9.430*** (0.000)	32.540*** (0.000)	7.70e+30*** (0.000)
LREC	0.520 (0.606)	0.520 (0.606)	24.770*** (0.000)	3.50e+30*** (0.000)

Note: p-values are reported in parentheses. ***, and ** indicate significance at the 1% and 5% levels, respectively. Abbreviations: LCF = Load Capacity Factor; AI = Artificial Intelligence; NR = Natural Resources; GTI = Green Technological Innovation; FD = Financial Development; REC = Renewable Energy Consumption.

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Table 4. Results of slope heterogeneity tests examining cross-sectional variations in coefficient estimates.

Practices		Delta tilde ($\hat{\Delta}$)	Delta tilde $\hat{\Delta}_{Adj.}$
Pesaran and Yamagata, (2008)	Stats.	5.752***	6.604***
	P-value	(0.000)	(0.000)
Blomquist and Westerlund, (2013)	Stats.	2.358**	2.707***
	P-value	(0.018)	(0.007)

Note: ***, and ** indicate significance at the 1% and 5% levels, respectively. $\hat{\Delta}$ and $\hat{\Delta}_{Adj.}$ denote the Delta and adjusted Delta slope heterogeneity test statistics, respectively.

Table 5. Results of second-generation panel unit root tests (CIPS and CADF) used to evaluate the stationarity properties of the study variables.

Series	Intercept				Intercept and Trend			
	Level		First Difference		Level		First Difference	
	Coeff.	P-value	Coeff.	P-value	Coeff.	P-value	Coeff.	P-value
CADF panel unit root test								
LLCF	-0.880	0.965	-3.471***	0.000	-1.858	0.835	-3.596***	0.003
LAI	-0.971	0.948	-2.892**	0.010	-1.184	0.992	-3.078**	0.049
LNR	-0.537	0.994	-4.606***	0.000	-2.737	0.179	-5.067***	0.000
LGTI	-2.368	0.105	-3.208***	0.001	-2.216	0.580	-3.738***	0.001
LFD	-2.556*	0.051	-3.928***	0.000	-3.203**	0.027	-3.817***	0.001
LREC	-1.068	0.923	-3.684***	0.000	-2.315	0.496	-3.843***	0.000
CIPS panel unit root test								
LLCF	-1.285		-5.107***		-2.212		-5.027***	

LAI	-1.214	-4.211***		-1.274	-4.543***	
LNR	-0.891	-5.763***		-2.623	-5.902***	
LGTI	-2.695***	-5.991***		-2.822**	-6.420***	
LFD	-3.389***	-5.667***		-3.943***	-5.684***	
LREC	-0.617	-4.238***		-2.157	-4.247***	
Critical Values	1%	5%	10%	1%	5%	10%
	-2.57	-2.33	-2.21	-3.1	-2.86	-2.73

Note: ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Abbreviations: LCF = Load Capacity Factor; AI = Artificial Intelligence; NR = Natural Resources; GTI = Green Technological Innovation; FD = Financial Development; REC = Renewable Energy Consumption.

Table 6. Results of panel cointegration tests indicating the existence of long-run equilibrium relationships among the study variables.

Kao cointegration test		
Statistic name	Statistic	p-value
Modified Dickey–Fuller t	-5.0497***	0.0000
Dickey–Fuller t	-3.8486***	0.0001
Augmented Dickey–Fuller t	-4.1068***	0.0000
Unadjusted modified Dickey–Fuller t	-7.1922***	0.0000
Unadjusted Dickey–Fuller t	-4.2730***	0.0000
Pedroni test		
Modified Phillips–Perron t	1.5557*	0.0599
Phillips–Perron t	-9.1867***	0.0000
Augmented Dickey–Fuller t	-2.8307***	0.0023
Westerlund test		
Variance ratio	-1.4185*	0.0781

Note: Under the null hypothesis, no cointegration is present. Under the alternative hypothesis, all panels exhibit cointegration in the Kao and Pedroni tests, whereas only some panels are cointegrated according to the Westerlund test. *** and * indicate significance at the 1% and 10% levels, respectively.

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Table 7. The MMQR estimation results capturing the heterogeneous effects of artificial intelligence, natural resources and control variables across different quantiles of the load capacity factor.

	Coefficient	Std. Err.	z	P>z	[95% conf. interval]	
Location						
LAI	-0.270***	0.024	-11.100	0.000	-0.318	-0.223
LNR	0.316***	0.022	14.340	0.000	0.273	0.359
LGTI	0.166***	0.044	3.740	0.000	0.079	0.253
LFD	1.206***	0.201	5.990	0.000	0.812	1.601
LREC	-0.222***	0.076	2.930	0.003	0.371	-0.073
_cons	0.609	0.412	1.480	0.140	-0.199	1.416
Scale						
LAI	-0.019	0.015	-1.310	0.192	-0.048	0.010
LNR	0.007	0.013	0.560	0.573	-0.018	0.033
LGTI	0.046	0.027	1.740	0.082	-0.006	0.098
LFD	0.111	0.121	0.920	0.360	-0.126	0.347
LREC	0.088*	0.046	1.940	0.053	-0.001	0.177
_cons	-0.232	0.247	-0.940	0.348	-0.716	0.252

qtile_10						
LAI	-0.240***	0.028	-8.540	0.000	-0.296	-0.185
LNR	0.305***	0.025	11.960	0.000	0.255	0.354
LGTI	0.093*	0.051	1.820	0.069	-0.007	0.194
LFD	1.033***	0.233	4.440	0.000	0.577	1.489
LREC	0.421***	0.088	4.784	0.000	0.533	-0.188
_cons	0.972**	0.476	2.040	0.041	0.039	1.906
qtile_25						
LAI	-0.253***	0.024	-10.380	0.000	-0.300	-0.205
LNR	0.309***	0.022	14.120	0.000	0.266	0.352
LGTI	0.123***	0.045	2.760	0.006	0.035	0.210
LFD	1.103***	0.201	5.500	0.000	0.710	1.496
LREC	0.501***	0.077	6.506	0.000	0.655	-0.155
_cons	0.825**	0.411	2.010	0.045	0.019	1.631
qtile_50						
LAI	-0.269***	0.024	-11.120	0.000	-0.317	-0.222
LNR	0.316***	0.022	14.440	0.000	0.273	0.359

LGTI	0.163***	0.044	3.700	0.000	0.077	0.250
LFD	1.201***	0.200	6.000	0.000	0.809	1.593
LREC	0.576***	0.076	7.578	0.000	0.677	-0.078
_cons	0.621	0.409	1.520	0.130	-0.182	1.423
qtile_75						
LAI	-0.284***	0.030	-9.610	0.000	-0.342	-0.226
LNR	0.322***	0.027	12.090	0.000	0.270	0.374
LGTI	0.200***	0.054	3.690	0.000	0.094	0.306
LFD	1.288***	0.244	5.290	0.000	0.810	1.765
LREC	0.628***	0.093	6.752	0.000	0.740	0.025
_cons	0.438	0.500	0.880	0.380	-0.541	1.417
qtile_90						
LAI	-0.305***	0.041	-7.490	0.000	-0.384	-0.225
LNR	0.330***	0.037	8.940	0.000	0.257	0.402
LGTI	0.249***	0.074	3.370	0.001	0.104	0.394
LFD	1.406***	0.337	4.170	0.000	0.746	2.066
LREC	0.764	0.127	6.015	0.616	0.812	0.185

_cons	0.191	0.689	0.280	0.781	-1.158	1.541
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Note: ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Abbreviations: AI = Artificial Intelligence; NR = Natural Resources; GTI = Green Technological Innovation; FD = Financial Development; REC = Renewable Energy Consumption.

Table 8. Pairwise Dumitrescu–Hurlin panel causality test results examining causal relationships among the study variables.

Null Hypothesis:	W-Stat.	Zbar-Stat.	Prob.	Remarks
LAI $\Leftrightarrow$ LLCF	3.4376***	2.8609	0.0042	Bidirectional causality
LLCF $\Leftrightarrow$ LAI	5.2613***	5.0804	0.0000	
LNR $\Leftrightarrow$ LLCF	5.1574***	4.9539	0.0000	Unidirectional causality
LLCF $\Leftrightarrow$ LNR	0.9369	-0.1826	0.8551	
LGTI $\Leftrightarrow$ LLCF	2.3265	1.5086	0.1314	No causality
LLCF $\Leftrightarrow$ LGTI	1.6405	0.6738	0.5005	
LFD $\Leftrightarrow$ LLCF	4.8411***	4.5689	0.0000	Bidirectional causality
LLCF $\Leftrightarrow$ LFD	2.8579**	2.1554	0.0311	
LREC $\Leftrightarrow$ LLCF	5.7261***	5.6460	0.0000	Unidirectional causality
LLCF $\Leftrightarrow$ LREC	0.6867	-0.4871	0.6262	
LNR $\Leftrightarrow$ LAI	4.4443***	4.0862	0.0000	Unidirectional causality
LAI $\Leftrightarrow$ LNR	0.6977	-0.4737	0.6357	
LGTI $\Leftrightarrow$ LAI	2.5461*	1.7758	0.0758	Bidirectional causality
LAI $\Leftrightarrow$ LGTI	3.1348**	2.4923	0.0127	

LFD $\Leftrightarrow$ LAI	3.1005**	2.4505	0.0143	
				Unidirectional causality
LAI $\Leftrightarrow$ LFD	1.0067	-0.0976	0.9223	
LREC $\Leftrightarrow$ LAI	4.5871***	4.2598	0.0000	
				Bidirectional causality
LAI $\Leftrightarrow$ LREC	2.9377**	2.2525	0.0243	
LGTI $\Leftrightarrow$ LNR	1.1491	0.0757	0.9397	
				No causality
LNR $\Leftrightarrow$ LGTI	1.5406	0.5521	0.5809	
LFD $\Leftrightarrow$ LNR	2.5908*	1.8303	0.0672	
				Unidirectional causality
LNR $\Leftrightarrow$ LFD	0.2125	-1.0644	0.2872	
LREC $\Leftrightarrow$ LNR	4.1256***	3.6982	0.0002	
				Unidirectional causality
LNR $\Leftrightarrow$ LREC	2.0776	1.2056	0.2280	
LFD $\Leftrightarrow$ LGTI	1.3145	0.27689	0.7819	
				No causality
LGTI $\Leftrightarrow$ LFD	0.4243	-0.80651	0.4199	
LREC $\Leftrightarrow$ LGTI	2.9158**	2.22578	0.0260	
				Unidirectional causality
LGTI $\Leftrightarrow$ LREC	1.5828	0.60351	0.5462	
LREC $\Leftrightarrow$ LFD	1.7026	0.74930	0.4537	
				No causality
LFD $\Leftrightarrow$ LREC	0.6636	-0.51525	0.6064	

Note: $\nRightarrow$ denotes “does not homogeneously cause”. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. Abbreviations: AI = Artificial Intelligence; NR = Natural Resources; GTI = Green Technological Innovation; FD = Financial Development; REC = Renewable Energy Consumption.

Highlights

- Significant cross-sectional dependency and slope heterogeneity observed among different countries.
- AI adoption has a statistically significant and negative impact on LCF.
- Natural resources exert a positive influence on LCF.
- Green technologies, financial development and renewable energy consistently increase LCF.
- Policy insights for achieving dual-carbon targets and advancing SDG-7 and SDG-13.

Relationship among AI, NR, GTI, FD, REC and LCF in Quad Nations

